

Akhilesh Basetty

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EDUCATION

UCLA, Bachelor of Science in Computer Science, 4.0 GPA

Class of 2028

- Relevant Coursework: Deep Learning, Robotics, Machine Learning, Computer Architecture, Programming Languages, Software Construction, Data Structures, Algorithms

Mission San Jose High School, 4.0 GPA, Valedictorian

Class of 2024

EXPERIENCE

Adobe

June – September 2026

Machine Learning Engineer Intern

- Built an LLM-driven Databricks pipeline that mines and clusters structural patterns from 5,000 production marketing journeys, providing insights on structural trends factors that boost performance with statistical significance testing.
- Enhanced Adobe's AI journey-generation agent with richer context by integrating mined patterns and marketing trends via confidence-gated retrieval layer, grounding generated journeys in real historical performance data.
- Generated journey fragment suggestions via embedding-based similarity search, ranked by performance lift and contextual relevance, for marketers to use directly in the journey canvas.

UCLA Kao Lab

March 2026 – Current

Undergraduate Researcher

- Built control software for ALOHA bimanual robot arms spanning low-level actuation and learned policy execution.
- Developed, fine-tuned, and deployed transformer-based robot policies, including Vision-Language-Action and imitation-learning flow matching approaches for manipulation tasks.
- Designed data collection pipelines and DAGger workflows to improve policy performance on real-world robotic tasks including laundry folding.

Spillbox

June – September 2025

Software Engineer Intern

- Designed and implemented distributed training pipelines for large-scale ML models using PyTorch DDP, enabling seamless scaling across GPU cloud instances.
- Engineered cloud bursting workflows by automating custom Kubernetes infrastructure deployment across AWS and Lambda Cloud, and enabled real-time syncing of training data and models between on-prem and cloud machines.
- [Demonstrated](#) end-to-end system by training a 1.5GB sentiment classifier in 7 minutes on a 2-GPU cluster with 50MB of training data, showcasing significant acceleration and resource efficiency.

UCLA Student Media

October 2024 – June 2026

DevOps Project Manager

- Developing and leading the BruinLife Photo Studio platform (Flask, React, SQL, AWS S3) for appointment booking, photo transactions, and photographer schedule management for 1,000+ annual customers.
- Managing Digital Ocean Droplets to maintain deployed media sites like Bruinwalk, BruinLife and other publications, enabling the 2,000+ organization to publish journalism.

PROJECTS

Nova, Tech for Good

October 2024 – Current

Project Lead, Finance Director

- Lead developer team to create applications for multiple local LA-based nonprofits, working with clients to deliver impactful solutions; contributed to the project's selection for the UCLA Engineering Society Technical Project Award.
- Built inventory management software for Mending Kids and an internal operations platform for Inner City Visions, supporting supply tracking and service delivery across international medical missions and at-risk youth programs.
- Secured funding through grants, sponsorships, and alumni/external partnerships.

UCLA American Society of Mechanical Engineers, X1 Robotics

October 2024 – June 2025

Software and Controls Lead

- Designed control systems to control a quadrupedal bear robot with limb reverse kinematics, motor PID and Feedforward control, and closed-loop gait trajectories, using ROS and MuJoCo.
- Integrated computer vision pipelines to identify faces and hand gestures and interact with people.
- Used Intel Realsense depth camera to avoid obstacles and follow paths, and IMU for closed loop kinematics.

SKILLS

- **Programming Languages:** Java, C++, Python, Javascript, HTML, CSS, Assembly, Bash, Lisp
- **Software Tools:** OpenCV, PyTorch, TensorFlow, ROS2, Node.js, React, Flask, SQLite, Firebase, MongoDB, S3, EC2

Awards: UCLA Engineering Scholarship (2025) | Presidential Scholars Candidate (2023) | National Merit \$2,500 Scholarship Recipient (2024)